

BHARTH K S

Robotics Engineer & AI Enthusiast | Generative AI & LLM Applications | Responsible AI | Cloud Integration
Edinburgh, UK | <mailto:bharth.ks2003@gmail.com> | Open to Hybrid – UK |
Portfolio - <https://portfolio-seven-gules-b3ntlu4hop.vercel.app/>

PROFESSIONAL SUMMARY

Motivated AI Engineer and MSc Robotics candidate at Heriot-Watt University with practical experience building Generative AI and LLM-based systems. Hands-on work spans adversarial AI safety research using multi-LLM architectures (Ollama, Anthropic Claude, Hugging Face), LLM fine-tuning, prompt engineering, and cloud-connected IoT data pipelines. Proficient in Python with a strong software engineering foundation, and experienced in integrating AI services via APIs to build intelligent, agent-driven applications. Committed to responsible AI principles including safety, auditability, and hallucination mitigation — directly aligned with regulated deployment environments.

CORE COMPETENCIES

Generative AI & LLMs: LLM Integration (Anthropic Claude, Ollama, OpenAI API), Prompt Engineering, System Prompts, Multi-LLM Architectures, RAG Patterns

AI Safety & Responsible AI: Adversarial Attack & Defense Systems, Hallucination Mitigation, Prompt Safety, AI Governance Principles

ML & Model Development: LLM Fine-Tuning (Hugging Face), Computer Vision (OpenCV), Machine Learning, Sensor Fusion

Cloud & API Integration: Cloud IoT Pipelines (MQTT, NodeRED, ThingSpeak), REST API Integration, Cloud Data Engineering

Programming & Engineering: Python (production-grade), C++, MATLAB, Data Structures & Algorithms, Git, VS Code

Frameworks & Tools: AI2-THOR, Hugging Face, Ollama, ROS 1 & 2, Jupyter, Docker (familiar), Arduino IDE

Soft Skills: Cross-functional Collaboration, Technical Communication to Non-Technical Stakeholders, Fast-paced Environments

AI & GENERATIVE AI PROJECTS

Adversarial Safety in Embodied AI — LLM Attack & Defence System *Heriot-Watt University | 2025–Present*
Python · Anthropic Claude · Ollama · Hugging Face · AI2-THOR

- Designed and implemented an adversarial attack-and-defence framework for LLM-driven agents operating within AI2-THOR, a 3D interactive embodied AI simulation environment — directly addressing real-world AI safety and robustness challenges.
- Built multi-LLM architecture leveraging Anthropic Claude and locally-deployed models via Ollama, enabling adversarial prompt injection attacks and corresponding defence mechanisms to evaluate model robustness under adversarial conditions. Fine-tuned LLMs using Hugging Face Transformers to improve agent resilience against adversarial inputs, applying supervised fine-tuning techniques to specialise model behaviour for safety-critical scenarios.
- Developed prompt engineering and system prompt strategies to harden LLM outputs, directly applicable to production-grade AI systems requiring hallucination mitigation and prompt safety controls.
- Integrated OpenAI API (in progress) to benchmark adversarial robustness across multiple LLM providers, establishing comparative evaluation pipelines — methodology applicable to enterprise multi-model AI deployments. Outcomes directly relevant to Responsible AI, GDPR compliance, and auditability requirements in regulated environments such as insurance and financial services.

GenAI Integration with Robotic Systems — HRI Module *Heriot-Watt University | 2025–Present*
Python · LLMs · OpenCV · Marty Robot

- Integrated Large Language Models into physical robotic systems (Marty), enabling natural language understanding and context-aware behavioural responses — practical experience building LLM-based agent workflows.
- Applied prompt engineering techniques and system prompts to constrain and direct LLM outputs for real-world robotic tasks, mirroring the design of production AI copilots and decision-support tools.
- Implemented computer vision pipelines using OpenCV alongside LLM integration, demonstrating multi-modal AI system design.

Cloud-Connected IoT Data Pipeline(Environment Monitoring System) *Heriot-Watt University | 2025–Present*

Python · ESP32 · MQTT · NodeRED · Jupyter · Cloud IoT

- Built a full-stack IoT solution using ESP32 for real-time sensor data collection, transmitted securely to cloud via MQTT protocol with NodeRED for workflow orchestration and Jupyter for analytics.
- Designed end-to-end cloud data pipeline — from device-level data ingestion to cloud storage and visual dashboarding — demonstrating skills directly transferable to AI data preparation, vector database feeding, and cloud-hosted AI service integration.

Autonomous Navigation System — ROS 2 Robotics Algorithms

Heriot-Watt University | 2025–Present

Python · C++ · ROS 2 · SLAM · Kalman Filter

- Developed autonomous navigation system implementing SLAM for real-time mapping and Kalman Filter for noise reduction, producing production-quality Python/C++ code adhering to software engineering best practices.
- Benchmarked multiple path planning algorithms for collision-free trajectory generation, demonstrating systematic evaluation methodology applicable to AI model comparison and selection.

EDUCATION

Heriot-Watt University

Edinburgh, UK

MSc Robotics | Sept 2025 – Sept 2026 (Expected)

- Human-Robot Interaction: LLM integration, GenAI deployment, machine learning, and computer vision — core AI engineering competencies.
- Internet of Things: Cloud IoT pipelines, MQTT, NodeRED, Jupyter — cloud data engineering and API integration skills.
- Robotics Systems Science: SLAM, Bayes/Kalman/Particle filters, sensor fusion — advanced algorithmic engineering.
- Intelligent Robotics: Robot kinematics, multi-modal sensing, swarm robotics, neurorobotics.

Kongu Engineering College

Erode, India

B.E. Mechatronics Engineering (Upper First-Class) | Sept 2021 – Apr 2025

- Core modules: Machine Vision & Image Processing, Python, C/C++, IoT, Data Structures, Drone Technology, LabVIEW, CNC, Manipulators, Robo Simulations- FANUC, ABB.

WORK EXPERIENCE

Production Engineering Intern — Body Shop

Hyundai Motors Pvt. Ltd, Chennai | June–July 2025

- Automated quality assurance workflows validating robotics performance data across high-volume production lines (i10, Venue, i20, Creta), maintaining standards in a safety-critical regulated environment.
- Produced structured operational data reports (output per minute/hour metrics) — experience in data-driven reporting and operational analytics relevant to AI monitoring and logging.

Team Member — KFC

Edinburgh, UK | Dec 2025–Present

- Performing reliably in a fast-paced, high-pressure environment requiring rapid decision-making, clear communication, and team coordination.

TECHNICAL SKILLS

AI / LLM Tools: Anthropic Claude API, Ollama (local LLMs), Hugging Face (fine-tuning & inference), OpenAI API (in progress), AI2-THOR

AI Safety: Adversarial prompt injection, defence mechanisms, hallucination mitigation, prompt safety design

Programming: Python, C++, MATLAB, C

Cloud & IoT: MQTT, NodeRED, ThingSpeak, Cloud IoT Pipelines, ESP32/ESP8266

ML & Vision: Computer Vision (OpenCV), Machine Learning, LLM Fine-Tuning, Sensor Fusion

Robotics & Tools: ROS 1 & 2, SLAM, SolidWorks, Autodesk CAD, Ansys, LabVIEW, RoboStudio

EXTRACURRICULAR & LEADERSHIP

- Team Member — Robotics Makers Society, Heriot-Watt University (2025–2026)
- Secretary — The Robotic Society, Kongu Engineering College (2024–2025)
- Technical Member — The Robotics Club, Kongu Engineering College (2023–2024)